

QRC / ARCS: Research Plan & Paper Structure

Answering your 13 research questions with a concrete section-by-section paper plan

Prepared for	Akash Kishore — QRC / ARCS project (22ET49, DSCE / VTU)
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Companion document	“QRC / ARCS: Project Changes & Execution Plan” — the engineering/repo side of this same plan
Builds on	“Novelty, Publishability & QACT Improvement Assessment” and “Final Plan for a Rigorous Case-Study Paper” (both already delivered)

0. Manuscript Type: Settle This First

You asked to reframe the paper as a review paper. I want to push back on the label before anything else, because IEEE Access is explicit and specific about this exact failure mode.

IEEE Access requires you to declare a manuscript type at submission (default: “Research Article”; “Topical Review” / “Survey” exist as separate types). Their own guidance states that *original review articles and surveys are acceptable even without new data, but a true Review is judged on comprehensiveness of synthesis, not on original experiments*. Editorial screeners specifically flag — as the single most common desk-rejection pattern — manuscripts whose declared type doesn't match their actual content: e.g., a paper full of original experimental benchmarking labeled as a “Survey.” That is exactly what would happen if you declare this a Review while it's built around your own V3/V4/V4a/V4b/QACT experiments.

What you actually want — broad grounding in current methodologies, honest comparison to other papers' results, forward-looking discussion of applications — doesn't require the Review label. It requires a well-built related-work / comparative-evaluation section inside a Research Article, which is exactly what a strong applications-oriented paper looks like. This document is built around that: “Research Article,” with an unusually thorough state-of-the-art and comparative-results component woven through it.

Practical instruction: when you submit, select “Research Article” as the manuscript type. Everything else you asked for (current methodologies, comparisons, future aspects) is built into the outline below — you get the content you wanted without the type-mismatch risk.

1. Master Map — Your 13 Questions → Where Each Is Answered

#	Your question	Answered in
1	Define the problem statement / what makes this unique	§2
2, 5, 8	Current methodologies; is the approach actually used elsewhere; comparative positioning	§3
3, 4	Your approach and exact methodology	§4
6	Algorithm comparison across noise regimes	§5
7	Simulation environment details and justification	§6
9	Advantages, applications, future aspects	§7
10	How equipment/algorithm choices affect the outcome	§8
11	Drawbacks and limitations	§9
12	Gradient-in-observation vs. gradient-in-reward, reported properly (best case → per-seed → aggregate)	§10
13	Failure modes, with proof from the project itself	§11

Section 12 at the end pulls all of this into one linear paper outline you can hand to a co-author or drop straight into an IEEE Access template.

2. Problem Statement and What Makes This Distinct (Q1)

2.1 Problem statement

Reinforcement-learning-based calibration has already been demonstrated on real superconducting qubit hardware (Sivak et al., 2022; Baum et al., 2021; Ding et al., 2023). This work does not attempt to repeat that demonstration. Instead, working entirely in simulation on a classical LC-resonator analogue of the resonance-tracking problem central to qubit calibration, this study asks: which specific encodings of derivative/rate information in an RL agent's observation space are necessary and sufficient for reliable frequency tracking under different drift statistics; whether that benefit is separable from the reward function's own shaping terms; and how fragile a trained policy is to superficially-compatible but semantically-altered observation spaces when transferred across simulators.

2.2 What is NOT the source of distinctiveness (say this plainly in the paper)

- “RL beats PID at tracking a resonance” — already shown, on real qubits, years before this project (§3).
- “Adding a derivative/gradient feature to an RL observation helps” — textbook control-theory practice (state-derivative augmentation, the RL analogue of a “D” term).
- Using Qiskit to represent a qubit-like system — done more thoroughly, on real hardware, elsewhere.

2.3 What IS distinctive — the actual claim to defend

- **A combined methodological package, not any single ingredient.** No cited comparator runs all of: a 4-way single-variable ablation (V3/V4a/V4b/V4) on the exact form of a derivative feature, a confound-clearing test isolating observation-level from reward-level shaping, dual noise-regime testing (Gaussian vs. Ornstein–Uhlenbeck) with formal significance testing, and a diagnosed cross-simulator transfer failure — together, on one system.
- **A specific, checkable negative-information finding.** Naive derivative encodings (raw, sign-only) score below a no-derivative baseline; only a specific frequency-normalized form helps. This is a feature-engineering-sensitivity claim, not a “more information helps” claim — it is more specific than anything in the cited derivative-augmentation literature.
- **A generalizable, diagnosed transfer-failure mechanism.** The sim-to-real/sim-to-sim literature (Tobin 2017, Zhao 2020) documents that transfer is fragile; it does not document this specific mechanism — a fixed-length observation vector that silently loads into a policy despite scrambled per-element semantics. That is a concrete, reproducible pitfall other RL-control practitioners can check for in their own pipelines.
- **An accessible, fully reproducible testbed.** Every comparator that touches real hardware requires qubit access most readers don't have. This testbed reproduces the essential epistemic constraint of qubit calibration — indirect, amplitude-only observation of a resonance — on a ~\$100 LC circuit and free software.

3. Current Methodologies, Real-World Adoption, and Comparative Positioning (Q2, Q5, Q8)

3.1 State-of-the-art table

Work	Venue / Year	Method	Adoption status / advancement	How this project compares
Fösel, Tighineanu, Weiss, Marquardt	PRX 8, 031084 (2018)	Neural-net RL for quantum feedback	Foundational; established the subfield is real and productive	Cite as origin point, not a target to outperform
Sivak et al.	PRX 12, 011059 (2022)	Model-free RL, real superconducting qubit	In active use — real-hardware deployment, ongoing line of work	This project trades real-hardware validity for full reproducibility and depth of ablation
Baum et al.	PRX Quantum 2, 040324 (2021)	Deep RL gate-set design, real hardware	In active use	Same trade-off as above
Ding et al.	arXiv:2304.06087 (2023)	PPO (same algorithm) for real fluxonium/transmon gate calibration	In active use, PPO specifically validated on hardware	Directly pre-empts “why PPO, why not real hardware” — cite explicitly
Porotti et al.	Quantum 3, 173 (2019)	Deep RL, quantum state prep, simulation-only	Closest simulation-only comparator	This project exceeds it on ablation depth, multi-seed reporting, a classical baseline, dual-noise-regime testing, and formal statistical testing — state this explicitly with a side-by-side table (see §3.2)
Tobin et al.	IROS (2017)	Domain randomization for sim-to-real	Standard practice in robotics RL	Frames why this project's Finding 3 (semantic-misalignment transfer failure) is a specific instance of a known class of problem
Zhao, Queralta, Westerlund	IEEE SSCI (2020)	Sim-to-real survey	Establishes the field	Same purpose as above

3.2 The one gap you need to close yourself

I do not have Porotti et al.'s exact quantitative benchmark numbers (fidelity, error, or their equivalent of MAE) reliably enough to build a numeric side-by-side table without risking a fabricated figure. Pull 2–3 concrete quantitative results directly from that paper (and, if feasible, Fösel 2018) and build a table shaped like:

Metric	Porotti et al. (2019)	This work
Baseline comparison used	[pull from paper]	PID, tuned per-regime
Number of seeds reported	[pull from paper]	5 (V4), extend to V4a/V4b/confound runs
Statistical test used	[pull from paper]	Mann-Whitney U, matched 1000-episode protocol
Noise regimes tested	[pull from paper]	Gaussian + Ornstein–Uhlenbeck

3.3 Is the specific technique (derivative/gradient observation augmentation) actually used elsewhere?

Yes, broadly, under different names — state this honestly rather than let a reviewer point it out. State-derivative or velocity augmentation is routine in continuous-control RL generally (it is the RL-observation analogue of a PID “D” term). The advancement here is not that it's used, but the specific empirical finding about which encodings of it help vs. actively hurt (§10), which the general derivative-augmentation literature does not resolve, because it doesn't typically test naive/malformed encodings against a no-derivative baseline the way the V4a/V4b comparison does.

4. Your Approach and Exact Methodology (Q3, Q4)

Component	Specification
Physical model	Classical damped LC resonator; Lorentzian amplitude response $A(f) = 1/(1+Q^2((f-f_0)/f_0)^2)$. See §6 for the f_0/Q values used and a discrepancy that needs resolving before this goes in the paper.
Drift models	Gaussian random-walk drift (used historically for V1–V4) and Ornstein–Uhlenbeck mean-reverting drift with 2% spike noise (currently active model), drift clipped to a bounded range around f_0 .
Baseline controller	PID, gains grid-searched per regime (seed and fix this search — see companion document §3).
RL algorithm	PPO (Stable-Baselines3 2.2.1), [256,256] MLP policy, 16 parallel envs, 2M steps, drift curriculum.
Observation variants under test	V3: 4-element obs, no derivative term. V4: 5-element obs adding a frequency-normalized gradient term $\Delta\text{amp} \cdot 3000/(\Delta\text{freq}+1)$, clipped to ± 1 . V4a: same but raw (unnormalized) Δamp . V4b: same but $\text{sign}(\Delta\text{amp})$ only (1-bit).
Reward (classical env)	$\text{ema_amp} + 0.5 \cdot \Delta\text{amp} \cdot \text{sign}(\text{action}) - 0.01 \text{action} + \text{low-amplitude penalty} + 3\text{-step dead-measurement termination penalty}$.
Evaluation protocol	1000-episode matched rollouts for each compared model, Mann-Whitney U test on per-episode MAE; 5 seeds per variant, report mean $\pm$ std across seeds as primary, not a single best seed (see §10 for the exact reporting template).
Cross-simulator transfer (QACT)	Same policy family, transferred into a Qiskit Aer single-qubit (Rx-rotation) circuit whose $P(1\rangle)$ plays the role of amplitude; evaluated zero-shot and after a small fine-tuning budget, once the alignment fixes in the companion document §6 are applied.

5. Algorithm Comparison Across Noise Regimes (Q6)

Use this exact table shape for the paper — it is already fully populated from existing results:

Controller	Gaussian MAE (Hz)	Gaussian near-res.	OU MAE (Hz)	OU near-res.	Gaussian vs. PID significance
PID (tuned)	3600 $\pm$ 5241	83.1% $\pm$ 18.5%	1414 $\pm$ 262	—	— (baseline)
V3 (no derivative)	3333 $\pm$ 1932	64.5%	1241 $\pm$ 254	97.7–97.9%	vs. PID: comparable mean, $\sim 1/10$ th the variance
V4a (raw Δamp)	4007 $\pm$ 2311	55.2%	— (recommend running)	—	Worse than V3 — include this row prominently
V4b (sign-only)	3793 $\pm$ 2051	55.4%	— (recommend running)	—	Worse than V3 — include this row prominently
V4 (5-seed aggregate)	2205 $\pm$ 518	84.8% $\pm$ 7.6%	1284 $\pm$ 93	97–98%	$p < 0.001$ (Gaussian); $p \approx 1.0$ (OU, not significant)

Two things to add before this is submission-ready: run V4a/V4b under the OU regime too (currently only reported under Gaussian, which leaves an asymmetry a reviewer will notice), and replace single-run V4a/V4b numbers with 5-seed aggregates matching V4's protocol.

6. Simulation Environment: Details and Justification (Q7)

A methods section needs to justify every parameter, not just state it. Here's what's traceable from the project so far, and one real inconsistency to resolve first:

f0 mismatch to resolve before writing: the RL simulation environment assumes $f_0 = 500$ kHz (drift clipped to 475–525 kHz), but the LTSpice-validated physical circuit resonates at $f_0 = 555.61$ kHz (Stage 3, confirmed in your original status report). That's a ~10% discrepancy between the simulated control problem and the circuit it's supposed to model. Decide and document one of: (a) update the RL environment's f_0 to 555.61 kHz and retrain/re-evaluate, so the simulation matches the as-built circuit; or (b) keep 500 kHz as a deliberate nominal design target and explicitly frame the 555.61 kHz SPICE result as realistic component-tolerance deviation the policy would have to handle — which is a defensible and even interesting framing, but only if you say so on purpose rather than leaving it as an undocumented mismatch a reviewer finds first.

Parameter	Value used	Justification needed for the paper
f_0 (resonant frequency)	500 kHz (RL env) vs. 555.61 kHz (SPICE)	Resolve the mismatch above; then state why this operating frequency (component availability? target application band?)
Q (quality factor)	50	Tie to actual component values (inductor SRR1280-100M, resonator capacitor, parasitic resistance) — show the $Q = f_0/\Delta f$ derivation from the LTSpice Bode plot, not just an assumed number
Drift models	Gaussian random walk (V1–V4 historical); Ornstein–Uhlenbeck + 2% spike noise (current)	Justify physically: what real phenomenon does each represent (e.g., thermal drift = slow random walk; environmental EMI = spike noise; flux noise analogue = mean-reverting OU)?
Drift clip range	475–525 kHz	State as $\pm 5\%$ of nominal — tie to a real constraint (varactor tuning range, or acceptable calibration-loop capture range)
Noise floor	0.02 (amplitude units)	Tie to a real SNR figure from the envelope-detector/ADC chain, not an arbitrary constant
Control cycle rate	Implied ~50 ms per your reward/EMA structure	State explicitly and justify against the RC low-pass filter's settling time (this is exactly what the Multisim/Simulink front-end validation in the companion document is for)

7. Advantages, Applications, and Future Aspects (Q9)

7.1 How the results can be used

- The V4a/V4b/V4 ablation table is directly reusable guidance for anyone adding derivative/rate features to an RL observation space in a tracking-control task — it's a checklist item (“normalize before you add it”), not just a result about this resonator.
- The diagnosed transfer-failure mechanism (§2.3) is a reusable pre-deployment check: before trusting a policy transferred across simulators or from simulation to hardware, verify observation-vector semantics element-by-element, not just dimensionality.

7.2 Where this generalizes beyond qubit calibration

- Any resonance/frequency-lock system with indirect (amplitude-only) feedback: PLL-based frequency synthesis, laser frequency stabilization, resonant wireless power transfer tuning, MEMS resonator drive-frequency tracking, antenna auto-tuning/impedance matching.

- More broadly: any RL-based control task where a derivative/rate observation feature is being considered as an engineering addition, in domains well outside quantum computing.

7.3 Future work (be specific, not generic)

- Recurrent policy (LSTM/GRU-based PPO) to test whether the OU-regime null result changes when the policy can carry state across steps, since OU drift has memory a feedforward policy structurally can't exploit.
- Extend the drift-rate robustness test to 750/1000 Hz/step and generate sample-efficiency curves — both already identified as pending in your own project notes and directly strengthen §5/§8.
- The equipment/algorithm sensitivity study proposed in §8 as a natural next paper or an extended results section.

8. How Equipment and Algorithm Choices Affect the Outcome (Q10)

You already have two concrete, real examples of this inside the project — use them as evidence rather than only arguing it in the abstract:

Example	What happened	What it demonstrates
f0 mismatch (§6)	RL environment assumes 500 kHz; the physically-validated (SPICE) circuit resonates at 555.61 kHz	A ~10% “equipment” (as-built circuit) vs. “algorithm-environment assumption” mismatch existed silently until this review — concrete proof that unreconciled parameters between the physical model and the RL environment directly affect what the reported results actually mean
DAC vs. DDS drift in the firmware BOM	The original status report specifies an MCP4921 DAC (12-bit SPI) for Stage 2 excitation; the current firmware (esp32_controller.ino) drives an AD9834 DDS chip instead — the ADC (MCP3202) stayed the same	A hardware-component decision changed after the RL/paper documentation was written and was not back-propagated — direct evidence that equipment choices and algorithm/paper assumptions can silently diverge over a project's lifetime

Recommended additional, structured study (new, not yet run) — a two-axis sensitivity analysis, entirely software-only:

Axis	Parameters to sweep	What it would show
“Equipment” axis (physical/simulated plant parameters)	Q-factor (e.g. 30/50/80), drift rate, noise floor, DAC/ADC bit-depth (quantization, via the virtual-HIL layer in the companion document)	How much of V4's advantage over PID survives changes in the physical system, independent of the RL algorithm
“Algorithm” axis (RL/training hyperparameters)	Entropy coefficient, learning-rate schedule, gradient-feature normalization constant	Whether the V4a/V4b finding (normalization matters) is specific to the 3000 Hz scale used, or generalizes across the hyperparameter space
Combined	A small grid across both axes for V4 only (not the full ablation matrix — keep this tractable)	Whether equipment sensitivity and algorithm sensitivity interact, which is a genuinely interesting, checkable finding for the discussion section

9. Limitations (Q11)

- Single scalar observation, known analytic response curve, 200-step episodes — a small control problem; the contribution is methodological rigor, not algorithmic difficulty.
- No physical hardware validation anywhere in this study — stated once, plainly, in a limitations section (see companion document §9 on what Multisim/Simulink can and cannot substitute for).
- The Qiskit “quantum” environment is a single-qubit Rx-rotation analogue — no entanglement, no multi-gate circuits, and (until fixed) no decoherence/readout noise model. Say this plainly rather than let “Qiskit” imply more physical fidelity than is actually modeled.
- Only PPO is tested — no comparison to other RL algorithms (SAC, TD3, DDPG); the derivative-encoding finding may or may not generalize across algorithm families.
- Seed variance is large for the core ablation (individual V4 seeds range from ~1510 to ~2887 Hz); report this honestly as a limitation of single-seed claims in PPO generally, not specific to this environment.
- The f0 mismatch and DAC/DDS documentation drift (§8) show the project's simulated environment and its physical-design documentation were not always kept in lockstep — disclose the resolution chosen, not just the fix.

10. Exact Reporting Protocol for the Gradient Feature (Q12)

Your instruction was explicit: don't just report MAE, isolate the observation-vs-reward confound, and show best case, per-seed, and full aggregate — clearly and in that order. Here is the exact structure to use, as a template you fill in once every rerun (companion document §1–§3) is complete:

Step 1 — Best-observed case (context, not the headline)

“The best-performing individual seed (Seed 3) achieved MAE 1510 ± 367 Hz, 94.8% near-resonance, 0.4% low-amplitude, under Gaussian drift.” State this once, explicitly labeled as the best of five, never as ‘the’ result.

Step 2 — Full per-seed table (not folded into the aggregate)

Seed	MAE (Hz)	Near-resonance	Low-amplitude
0	[fill in — in the 2288–2887 Hz range per existing notes]	[fill in]	[fill in]
1	1731	[fill in]	[fill in]
2	[fill in]	[fill in]	[fill in]
3	1510	94.8%	0.4%
4	[fill in]	[fill in]	[fill in]

Step 3 — Aggregate (the actual headline)

“Across all 5 seeds, V4 achieves MAE 2205 ± 518 Hz, 84.8% ± 7.6% near-resonance, 5.1% ± 3.7% low-amplitude under Gaussian drift (Mann-Whitney U vs. V3, $p < 0.001$).” This is the number every other comparison in the paper should reference.

Step 4 — Observation-vs-reward confound isolation (run once, report regardless of outcome)

Variant	Observation	Reward	Purpose
V4 (existing)	Has gradient term	Has gradient bonus	Combined effect — already run
New run A	Has gradient term	Gradient bonus removed	Isolates: does the observation feature help on its own?
New run B	No gradient term (V3-style)	Has gradient bonus	Isolates: does the reward shaping alone explain the V3→V4 gap?

Report all three (V4, Run A, Run B) side by side with the same 5-seed, Mann-Whitney protocol as Step 3 — whichever way it comes out, this is the single most convincing piece of methodological rigor available to you, because it directly tests the paper's own central claim rather than assuming it.

11. Failure Modes — With Proof From This Project (Q13)

Failure mode	Proof (from this project)	Effect on results	Status
Best-of-N seed reported as “the” result	Seed 3 (MAE 1510 Hz) reported alone in the original status report; 5-seed aggregate is 2205 ± 518 Hz	Headline number materially overstated versus the honest aggregate	Fix: report aggregate as primary everywhere (§10)

Failure mode	Proof (from this project)	Effect on results	Status
Silent observation-vector semantic misalignment across simulators	Classical obs order [norm_freq, ema_amp, prev_ema_amp, prev_action, amp_gradient] vs. Qiskit obs order [amp, freq_err_norm, prev_action, delta_amp, amp_grad] — same length (5), SB3 loads the mismatched checkpoint without error	Zero-shot QACT MAE inflated to 2964 ± 2473 Hz / 32.8% near-resonance — mostly a code bug, not quantum-domain difficulty	Isolation test proved the fix works ($\rightarrow 1708 \pm 1103$ Hz / 66.5%); full realignment pending (companion document §6)
Unseeded stochastic evaluation	Independent rerun of QACT v4/v5 gave 2732 ± 1839 / 3670 ± 6128 Hz vs. saved 2176 / 3237 ± 4581 Hz; PID grid search gains changed between runs ($1.0/0.01/0.05 \rightarrow 1.0/0.005/0.05$)	Currently saved QACT numbers are not reproducible and cannot be cited as-is	Fix: seed every RNG (AerSimulator, PID search, env resets) before any number goes in the paper
Dead code paths / broken imports	phase1_baseline.py and sweep_verify_v2.py import a qiskit_integration.qiskit_resonator_env_v2 module that does not exist in the repo (only _v3 present)	These scripts cannot currently be run at all	Fix: restore/recreate the module or reprint the scripts
Hardcoded absolute paths surviving a project relocation	Qiskit-integration scripts reference C:\MiniProject_Sem4\... while the project now lives at a different path	Breaks the pipeline on the current machine independent of any code logic issue	Fix: relative/config-based paths (companion document §4)
Reward/observation confound in the core ablation	V4's reward already contains a gradient-shaped term $(0.5 \cdot (\text{ema_amp} - \text{prev_ema_amp}) \cdot \text{sign}(\text{action}))$; the observation's gradient feature partially duplicates it	The V3 $\rightarrow$ V4 causal claim (“the observation feature helps”) is not yet cleanly isolated from reward shaping	Fix: run the two confound-isolating variants in §10, Step 4
Reward-aligned evaluation metrics	Near-resonance (>0.90) and low-amplitude (<0.70) thresholds are defined directly in terms of the reward's own ema_amp signal, and termination is defined the same way	“94.8% near-resonance” is partly a statement about reward alignment, not an independent tracking-quality measurement	Fix: lead with MAE vs. true f0 as the primary, independent metric; keep near/low as secondary
Metric strictness mismatch between simulators	Qiskit env's Rabi curve ($\omega_{\text{rabi}}=2500$) crosses the 0.90 near-resonance threshold at ± 1906 Hz vs. the classical Lorentzian's ± 3333 Hz — $\sim 1.75\times$ stricter for nominally the same threshold	Part of the apparent QACT “transfer gap” is a metric-definition artifact, not a performance difference	Fix: match $\omega_{\text{rabi}} \approx 5000$ Hz, or report a width-normalized metric and disclose the asymmetry
Undocumented parameter drift between physical design and simulation	RL environment's $f_0 = 500$ kHz vs. SPICE-validated circuit's $f_0 = 555.61$ kHz; original BOM specifies an MCP4921 DAC while the current firmware drives an AD9834 DDS	The simulated control problem may not correspond to the as-built (or as-documented) physical system	Decide and document explicitly (§6, §8) before writing the methods section

12. Final Paper Outline

Section	Contents
Abstract	State the narrowed thesis (§2.1) and the three headline findings (feature-encoding sensitivity, confound isolation result, diagnosed transfer mechanism) in ~150 words, with explicit numbers from the 5-seed aggregate, not Seed 3.
1. Introduction	Motivate via real-hardware RL calibration literature (§3.1); state plainly this work doesn't attempt to replicate that on real qubits; state the problem statement (§2.1).
2. Related Work	The §3.1 table, organized as: real-hardware RL-for-calibration (context, not a target to beat) vs. Porotti (a target you concretely compare against, §3.2) vs. sim-to-real/sim-to-sim transfer literature (frames §11's transfer-mechanism finding).
3. Methodology	§4 in full — physical model, drift models, PID baseline, PPO setup, all four observation variants, evaluation protocol. Include the resolved f0/DAC-DDS discussion from §6/§8 here, not buried in an appendix.
4. Results I — Feature-Encoding Ablation	The §5 noise-regime table and the §10 four-step reporting protocol (best case → per-seed → aggregate → confound isolation) in full.
5. Results II — Cross-Simulator Transfer	QACT setup, the misaligned-vs-aligned isolation-test table, and zero-shot/fine-tuned numbers once the companion document's §6–§8 fixes are applied and rerun.
6. Equipment/Algorithm Sensitivity	The two documented real examples plus the proposed sensitivity grid, §8.
7. Discussion	What generalizes beyond this testbed (§7.2); direct engagement with the failure-modes table (§11) as a set of transferable lessons, not just a list of things that went wrong.
8. Limitations and Future Work	§9 and §7.3 combined; state hardware-out-of-scope once, plainly.
9. Conclusion	Restate the three headline findings with their aggregate numbers.

This is a Research Article with an unusually thorough comparative and methodological backbone — which is what actually earns the “rigorous” description you're after, without the manuscript-type risk of calling it a Review.